

Nifty 100 Financial Intelligence Platform

Analyst Guide

A practical guide to the data foundation, analytics engine, dashboard, API, clustering, financial intelligence, reporting, testing, and source-truth principles of the platform.

Universe	Nifty 100 source dataset
Companies	92
Database	SQLite
Dashboard	Streamlit
API	FastAPI
Automated tests	168 passing
Company archetypes	5 KMeans clusters
Primary objective	Financial analytics and research automation

1. Platform Overview

The platform combines data engineering, financial analytics, screening, peer comparison, valuation, clustering, NLP-style financial signals, dashboards, APIs, and automated reporting.

- 92 companies are represented in the source company universe.
- Financial data is normalized and stored in SQLite.
- Financial ratios and growth metrics are calculated programmatically.
- A multi-factor screener supports both presets and custom filters.
- Peer analytics provide percentile-based relative comparison.
- A Streamlit dashboard provides interactive analyst workflows.
- A FastAPI service exposes the platform programmatically.
- KMeans clustering groups companies into five financial archetypes.
- Automated PDF reports summarize company and portfolio intelligence.

The project follows a source-truth-first design. Missing, unusual, or incomplete source information is documented rather than fabricated or adjusted only to satisfy expected outcomes.

2. Data Foundation

Sprint 1 established the data model, ETL workflow, normalization rules, SQLite database, and data-quality framework used by the rest of the platform.

- Raw data is organized into core and supplementary source folders.
- Company identifiers are normalized to ticker-based keys.
- Year values are normalized into consistent reporting periods.
- Duplicate financial rows are removed using deterministic rules.
- Invalid or orphan records are rejected before database loading.
- The SQLite database contains 12 analytical tables.
- Foreign-key integrity is validated after loading.
- A multi-rule data-quality engine records validation failures.

Companies	92
Profit and Loss rows	1,070
Balance Sheet rows	1,140
Cash Flow rows	1,056
Stock price rows	5,520
Market-cap rows	552
Peer-group rows	56
Foreign-key violations	0

Data-quality checks are treated as part of the analytical pipeline, not as a separate afterthought. Critical issues are resolved before downstream analytics are considered reliable.

3. Financial Ratio Engine

Sprint 2 created the reusable KPI layer that powers screening, peer analysis, valuation, clustering, dashboard views, and API responses.

- Profitability metrics include ROE, ROCE, ROA, NPM, and OPM.
- Leverage and efficiency metrics include Debt-to-Equity and related ratios.
- Revenue, PAT, and EPS CAGR calculations handle multiple edge cases.
- Financial-sector companies are handled separately where appropriate.
- Cash-flow conversion and growth metrics support quality analysis.
- Computed ratios are persisted for reuse across the platform.

Financial ratio rows	1,155
Primary use	Screening and company analysis
Growth handling	CAGR with edge-case flags
Sector-aware logic	Yes
Downstream reuse	Dashboard, API, peers, valuation, clustering

The ratio engine preserves unusual source-derived values rather than silently replacing them. This is important because some source ROE and ROCE observations are extreme and materially affect statistical analysis.

4. Screener and Peer Comparison

Sprint 3 added preset screening strategies, custom filtering, composite scores, peer-group analytics, and percentile-based comparison.

- The screener supports both preset strategies and custom filters.
- Preset logic is kept exact instead of being weakened to increase counts.
- Company rankings combine multiple financial dimensions.
- Peer groups allow like-for-like comparison within comparable businesses.
- Percentile ranks provide relative positioning across peer metrics.
- Radar-chart data is generated for visual peer comparison.

Quality Compounder	22 companies
Value Pick	2 companies
Growth Accelerator	19 companies
Dividend Champion	30 companies
Debt-Free Blue Chip	2 companies
Turnaround Watch	31 companies
Peer groups	11
Peer percentile rows	560

The screener is intended as an analytical narrowing tool rather than a recommendation engine. A low count in a strict preset is considered valid when it reflects the actual rule set.

5. Dashboard and Valuation

Sprint 4 delivered the interactive analyst dashboard and valuation workflow.

- The Streamlit dashboard contains eight analyst-facing screens.
- Database access is cached to improve dashboard responsiveness.
- Company profiles combine fundamentals, ratios, valuation, and reports.
- The screener can be used interactively from the dashboard.
- Peer comparison and sector analysis are available visually.
- Capital-allocation data is integrated where source coverage exists.
- The valuation engine combines market-cap, earnings, and cash-flow metrics.

Day 22	Streamlit foundation and cached DB layer - DONE
Day 23	Home and Company Profile - DONE
Day 24	Screener and Peer Comparison - DONE
Day 25	Trends, Sectors, Capital Allocation and Reports - DONE
Day 26	Valuation Module - DONE
Day 27	Dashboard QA - DONE
Day 28	Documentation and final release - DONE
Sprint 4 Status	COMPLETE

The valuation module exports reusable analytical outputs and is also consumed by dashboard and reporting workflows.

6. Financial Intelligence and Automated Reporting

Sprint 5 expanded the platform from traditional financial analytics into rule-based intelligence, cash-flow analysis, capital-allocation analysis, and automated PDF reporting.

- Financial text and analysis data are parsed into structured information.
- A 24-rule engine generates evidence-based PRO and CON signals.
- Signal confidence is based on the underlying financial conditions.
- Cash-flow intelligence evaluates FCF quality, conversion, and growth.
- Distress and deleveraging indicators provide additional risk context.
- Capital-allocation patterns are analyzed from available source history.
- Company tear sheets are generated automatically with charts and KPIs.
- Peer-group reports and a portfolio summary provide broader context.

Signal rules	24
Generated PRO signals	398
Generated CON signals	108
Total signals	506
Companies with PRO coverage	89
Companies with CON coverage	57
Eligible company tear sheets	88
Peer-group reports	11

Coverage is deliberately not forced to 100 percent. A company receives a PRO or CON only when the relevant rule is satisfied. Likewise, companies with insufficient annual history are not given fabricated tear sheets.

7. Clustering and Statistical Intelligence

Sprint 6 introduces unsupervised learning and portfolio-level statistical analysis to identify financial archetypes, relationships, and outliers.

- Five financial features are standardized before clustering.
- Missing observations use sector-median imputation where possible.
- A portfolio median is used only when the sector itself has no usable value.
- KMeans uses five clusters with `random_state=42` for reproducibility.
- Elbow analysis evaluates candidate cluster counts from 2 through 10.
- A 10-KPI correlation matrix supports relationship analysis.
- Sector-relative z-scores identify extreme observations.
- Portfolio statistics provide distribution and percentile context.

Core Balanced Companies	58
High Margin / Mixed Quality	16
ROE Outliers	2
FCF Growth Outliers	2
Leveraged Financial Growth	14
Total clustered companies	92

Cluster names are descriptive labels derived from cluster profiles. They are analytical archetypes, not investment ratings.

8. FastAPI Service

Sprint 6 exposes the analytical platform through a versioned FastAPI service for programmatic access and integration.

- The API uses the /api/v1 prefix.
- A health endpoint reports service status and database counts.
- Company endpoints expose profiles and financial statements.
- Ratio history can be retrieved by company and reporting period.
- The screener is accessible through query parameters.
- Sector and peer-group endpoints support comparative analysis.
- Market-cap and portfolio-statistics endpoints are available.
- Company documents and tear sheets are accessible through the API.
- OpenAPI and Postman specifications are exported in docs.

API framework	FastAPI
Version prefix	/api/v1
API endpoints	16
Local port	8000
Interactive docs	/docs
Alternative docs	/redoc
Health endpoint	/api/v1/health
OpenAPI export	docs/openapi.json

The API reuses the analytical engines used elsewhere in the platform. This helps prevent calculation differences between the dashboard, API, and underlying analytics.

9. Testing, QA and Performance

Automated testing and performance validation protect the data pipeline, analytics, API behavior, and integration workflows.

- ETL tests validate normalization and loading behavior.
- KPI tests verify financial calculations and edge cases.
- Data-quality tests cover the validation framework.
- API tests cover companies, screening, sectors, peers, and documents.
- Integration tests compare API screening with the core engine.
- An HTML pytest report provides test-run evidence.
- Concurrent screener calls are tested for performance.
- Company-profile endpoints are tested for response time.
- Streamlit and FastAPI are verified to operate simultaneously.

Automated tests	168
Current failures	0
API tests	32
Black	PASS
Ruff	PASS
Screener target	Less than 10 seconds
Company profile target	Less than 3 seconds
HTML report	reports/pytest_report.html

The verified project regression suite currently completes with 168 passing tests and zero failures.

10. Analyst Workflow

The platform is designed to support a repeatable research workflow from universe selection through company-level investigation.

Step 1 - Screen the universe

Start with a preset strategy or custom screener conditions to reduce the company universe to candidates matching the required financial characteristics.

Step 2 - Review company fundamentals

Inspect historical profit and loss, balance-sheet, cash-flow, ratio, growth, and market-cap information.

Step 3 - Compare peers

Use peer percentiles and radar-style comparisons to understand whether a company is strong or weak relative to comparable businesses.

Step 4 - Evaluate valuation

Review valuation metrics together with profitability, growth, cash-flow quality, and historical context rather than treating valuation as a standalone decision.

Step 5 - Review intelligence signals

Use PRO and CON signals as evidence summaries. A missing signal does not imply a positive or negative conclusion.

Step 6 - Inspect statistical context

Review company clusters, portfolio percentiles, correlations, and outliers to understand broader positioning.

Step 7 - Review source documents

Where important decisions depend on a number or narrative, analysts should review the available source documents and independently verify material information.

11. Source-Truth Principles and Known Constraints

The platform deliberately distinguishes between a software defect and a limitation inherited from the source data.

- The source company universe contains 92 usable company records.
- The broad sector taxonomy contains 10 sectors while peer analytics contain 11 peer groups.
- Some source-derived ROE and ROCE observations are unusually large.
- Extreme observations are retained when they are traceable to source data.
- PRO and CON signals are not fabricated to force full company coverage.
- Current rule-based coverage includes 89 companies with at least one PRO and 57 with at least one CON.
- Only 88 companies currently meet the annual-history requirement for full company tear-sheet generation.
- AMBUJACEM, JIOFIN, NESTLEIND, and SIEMENS are skipped by the batch tear-sheet process when they do not satisfy the required annual history.
- ATGL has unavailable source information for parts of the capital-allocation and cash-flow intelligence workflow.
- Missing values are reported transparently rather than replaced with invented financial conclusions.

These constraints are important when interpreting acceptance criteria. The project prioritizes reproducibility and source truth over manipulating data merely to obtain a desired checklist result.

12. Operating Guide and Conclusion

The platform can be operated locally as an analytics application, dashboard, API service, and reporting system.

Run the dashboard

```
python -m streamlit run src/dashboard/app.py
```

Run the API

```
python -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000 --reload
```

Run clustering

```
python -m src.analytics.clustering
```

Run valuation

```
python -m src.analytics.valuation
```

Run automated tests

```
pytest -q
```

Run code-quality checks

```
python -m black --check src tests
```

```
python -m ruff check src tests
```

Company universe	92
Financial ratio records	1,155
Broad sectors	10
Peer groups	11
KMeans clusters	5
FastAPI endpoints	16
Dashboard screens	8
Automated tests	168 passing

The Nifty 100 Financial Intelligence Platform demonstrates an end-to-end financial analytics workflow combining data engineering, financial modelling, statistical analysis, machine learning, interactive visualization, APIs, automated testing, and reporting.

The system is intended for educational, analytical, and portfolio-development purposes. It does not provide investment advice.

13. Streamlit Screener - How to Use It

Start the dashboard from the project root with **streamlit run src/dashboard/app.py**. Open the local Streamlit URL shown in the terminal and navigate to the Screener screen.

Step	Action
1	Open the Screener page from the dashboard navigation.
2	Choose a built-in preset or configure custom filters.
3	Set filters such as minimum ROE, maximum debt-to-equity, minimum revenue CAGR, minimum PAT CAGR, sector, FCF and P/E.
4	Run the screener to generate the ranked company list.
5	Review metrics and composite ranking.
6	Export the filtered result when CSV download is required.
7	Cross-check shortlisted companies using Profile, Peers, Trends and Reports.

Built-in presets use their defined financial rules without weakening thresholds merely to increase the number of results.

14. Dashboard Navigation - All 8 Screens

1. Home

Provides the portfolio-level landing view with headline KPIs, coverage information and high-level financial summaries.

2. Company Profile

Displays one company's financial history, key ratios, latest metrics, business information and supporting analysis.

3. Screener

Filters and ranks companies using financial rules, presets and custom thresholds.

4. Peers

Compares companies within peer groups using percentile metrics, benchmarking and radar-style relative analysis.

5. Trends

Shows historical financial and ratio trends to identify growth, deterioration, consistency and cyclicity.

6. Sectors

Summarizes broad-sector performance and enables sector-level comparison of companies.

7. Capital

Focuses on capital allocation, cash-flow intelligence, leverage changes and related financial-quality signals.

8. Reports

Provides access to company tearsheets, sector reports and portfolio-level reports.

A typical workflow begins on Home, moves to Screener for discovery, uses Company Profile and Trends for historical analysis, Peers and Sectors for relative comparison, Capital for cash-flow analysis, and Reports for final review.

15. Generating PDF Tearsheets

Company tearsheets are generated from the SQLite financial database, calculated ratios and intelligence outputs. Run the commands from the project root while the virtual environment is active.

Single-company reporting module

```
python -m src.reports.tearsheet
```

Batch tearsheet generation

```
python -m src.reports.batch_tearsheets
```

Generated company PDFs are stored under **reports/tearsheets/**. Companies without enough annual financial history may be skipped according to the reporting eligibility rules.

The platform preserves source-data limitations rather than fabricating missing financial history.

16. FastAPI Usage and Example curl Commands

Start the REST API server from the project root:

```
python -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

Interactive OpenAPI documentation is available at <http://127.0.0.1:8000/docs> while the server is running.

Health check

```
curl "http://127.0.0.1:8000/api/v1/health"
```

List companies

```
curl "http://127.0.0.1:8000/api/v1/companies"
```

TCS company profile

```
curl "http://127.0.0.1:8000/api/v1/companies/TCS"
```

TCS ratios

```
curl "http://127.0.0.1:8000/api/v1/companies/TCS/ratios"
```

Screener example

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_de=1"
```

Sector summary

```
curl "http://127.0.0.1:8000/api/v1/sectors"
```

Portfolio statistics

```
curl "http://127.0.0.1:8000/api/v1/portfolio/stats"
```

TCS documents

```
curl "http://127.0.0.1:8000/api/v1/companies/TCS/documents"
```

17. Troubleshooting Common Issues

Issue	Recommended action
ModuleNotFoundError: src	Run commands from the project root and ensure the virtual environment is active.
API port 8000 already in use	Stop the existing API process or start uvicorn on another port.
Streamlit port 8501 already in use	Stop the earlier Streamlit process or use another available port.
Database or table not found	Confirm db/nifty100.db exists and ETL completed successfully.
Screener returns fewer companies than expected	Review active filters. Exact presets may legitimately return small result sets.
Missing ratios or CAGR values	Check source-history coverage and required calculation inputs.
Tearsheet not generated	Verify sufficient annual financial history and reporting inputs.
PDF rendering problem	Regenerate and visually inspect for clipping, overflow or broken charts.
Tests modify generated Excel outputs	Restore unintended tracked workbook changes with git restore.
Ruff or Black reports issues	Run Black, then Ruff, then rerun the full pytest suite.
Unexpected source counts	Verify source data before changing code. Document source/spec mismatches instead of fabricating records.

Disclaimer: This platform is an analytical and educational tool. Outputs should be verified against source financial statements and should not be treated as investment advice.